x_1	x_2	x_3	x_4	x_5	x_6
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

$$\begin{bmatrix}4\\-2\\-2\end{bmatrix}$$ *Initial W* **REPEAT using Step η = 1**

Check classes of $\begin{bmatrix}0\\0\end{bmatrix}\ \begin{bmatrix}7\\6\end{bmatrix}\ \begin{bmatrix}3\\2\end{bmatrix}$ (Z_1 Z_2 Z_3)

C_1 positive C_2 negative Zero is unaccepted

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
1	1	1	1	1	1
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 2

No change in W
Stop Iterations

$d(x_1) = \begin{bmatrix}1&1&1\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_2) = \begin{bmatrix}1&2&1\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_3) = \begin{bmatrix}1&1&2\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_4) = \begin{bmatrix}1&5&4\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_5) = \begin{bmatrix}1&6&5\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_6) = \begin{bmatrix}1&6&6\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$